

Loan Approval Prediction

Group Hedgers

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1. Introduction

- The global lending industry is valued at over **\$8 trillion**, with banks and financial institutions processing **millions of loan applications annually** (Statista, 2023).

Bias in Loan Approval

- Minority applicants are **40% more likely** to be denied loans than white applicants, even with similar profiles (HBR, 2021).
- **Solution:** ML models reduce human bias and promote fairer lending.

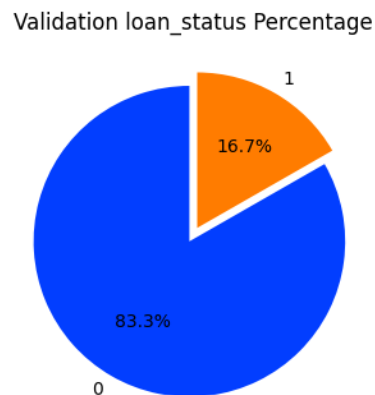
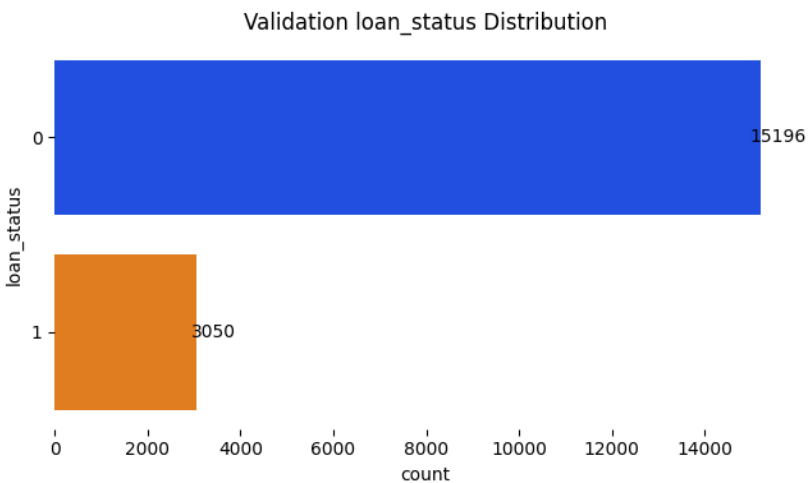
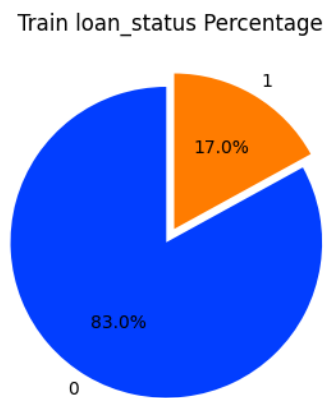
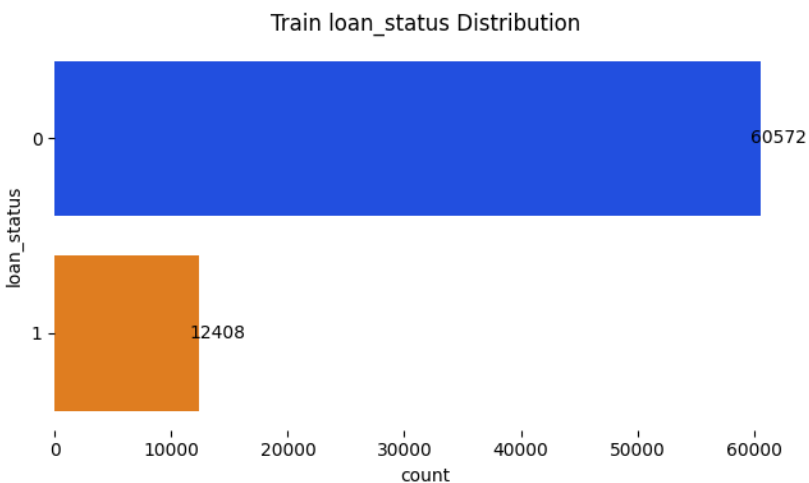
High Loan Default Rates

- Global loan default rates range from **2-5%**, small business loans exceed **10%** (Federal Reserve, 2023).
- **Solution:** AI models can **lower bad loan rates by 20%**.

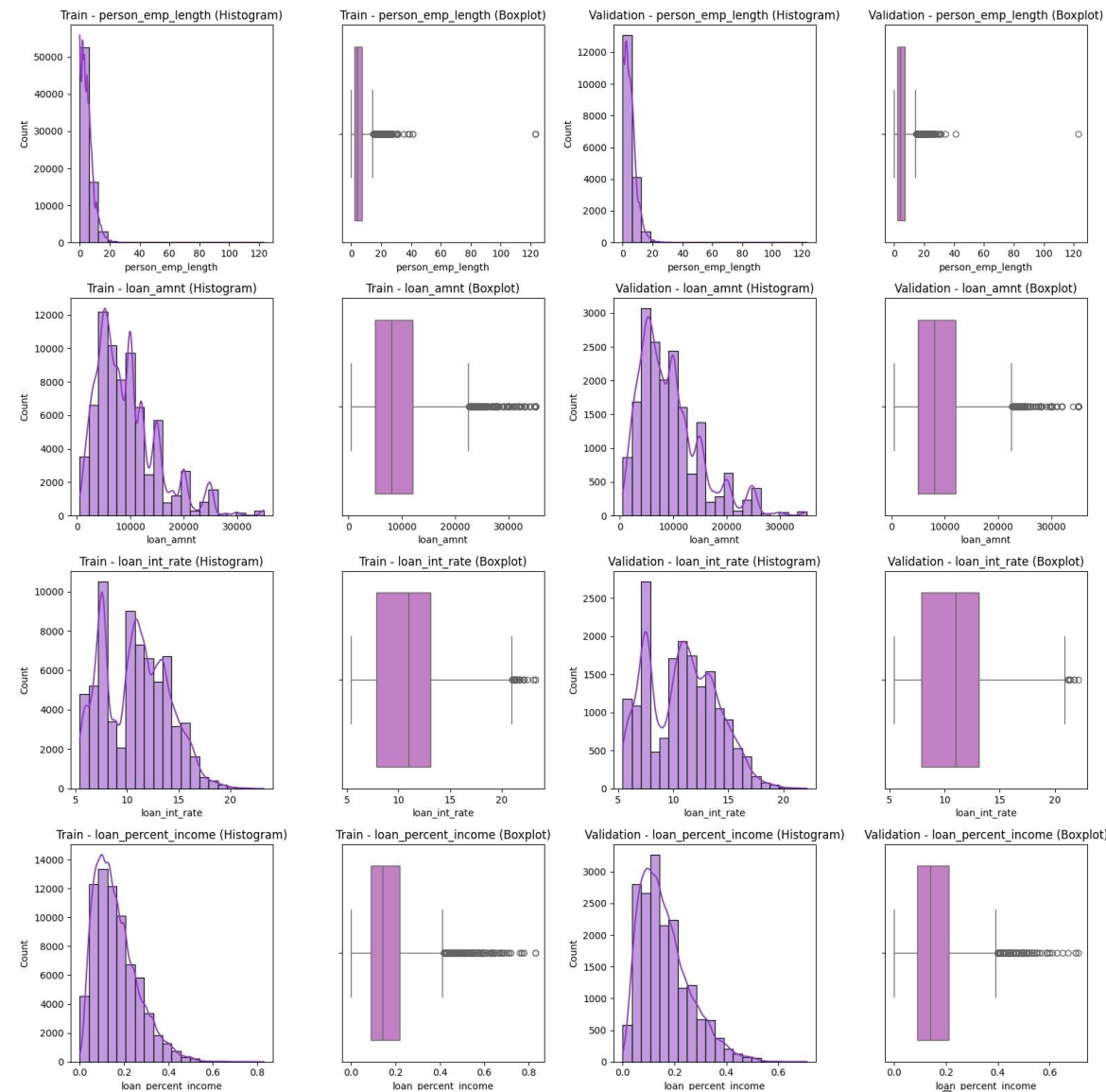
Slow Processing Time

- Traditional underwriting takes **5-7 days** for a single loan.
- **Solution:** Automated ML models are able to process approvals **in seconds**.

2. EDA

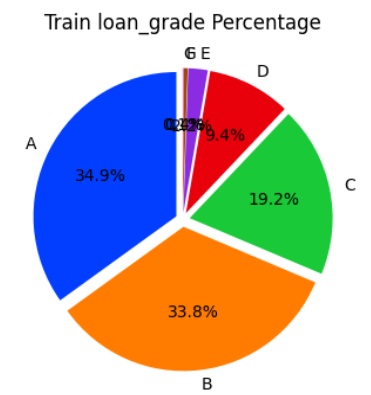
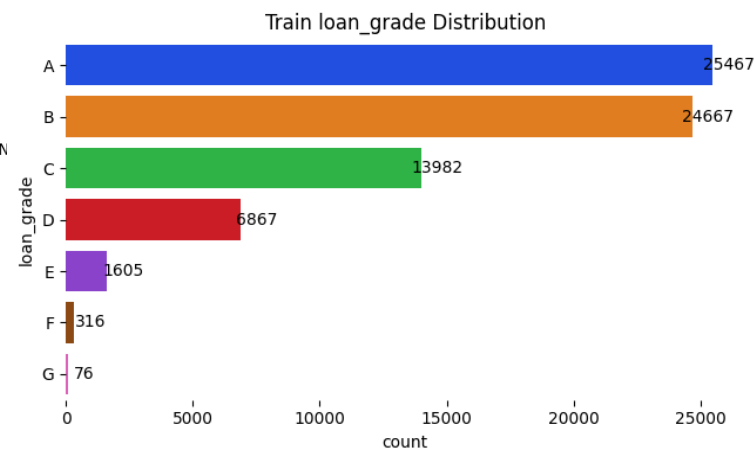
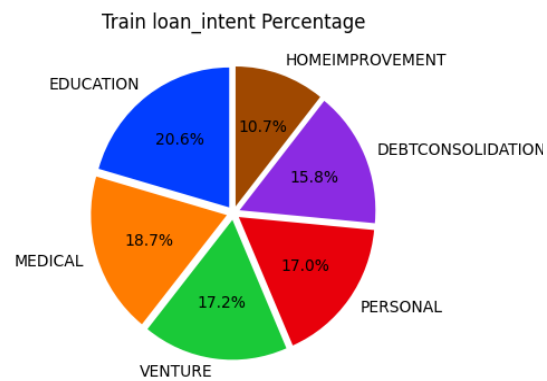
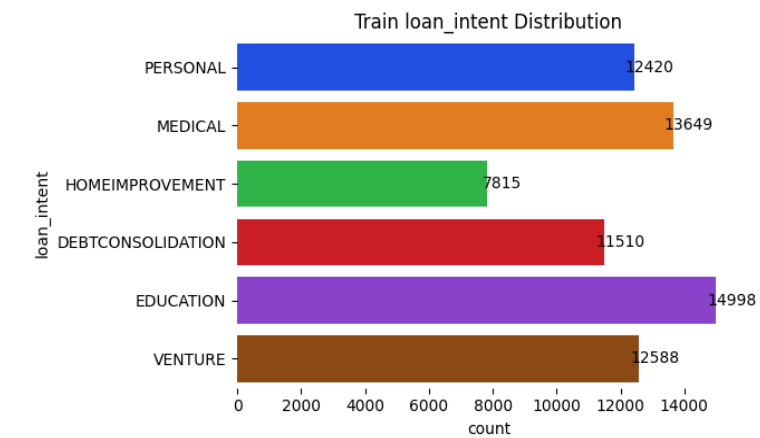
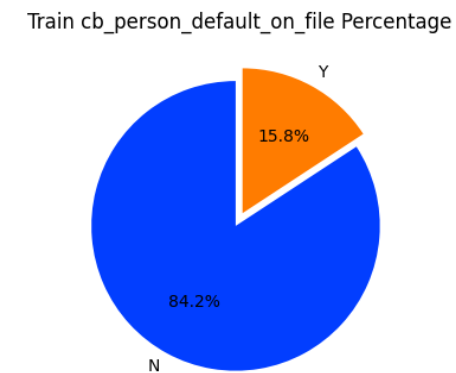
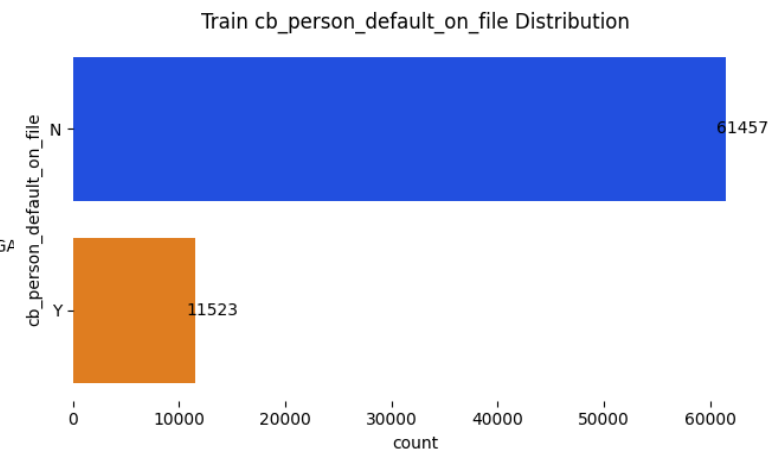
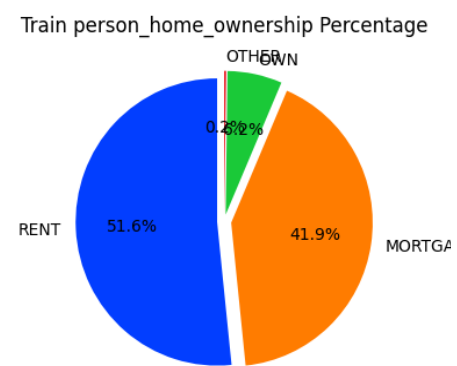
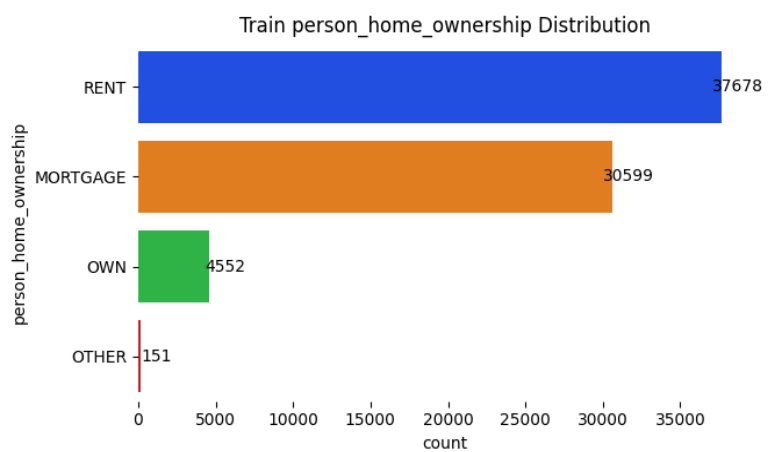


Distribution and percentage of target (loan status)



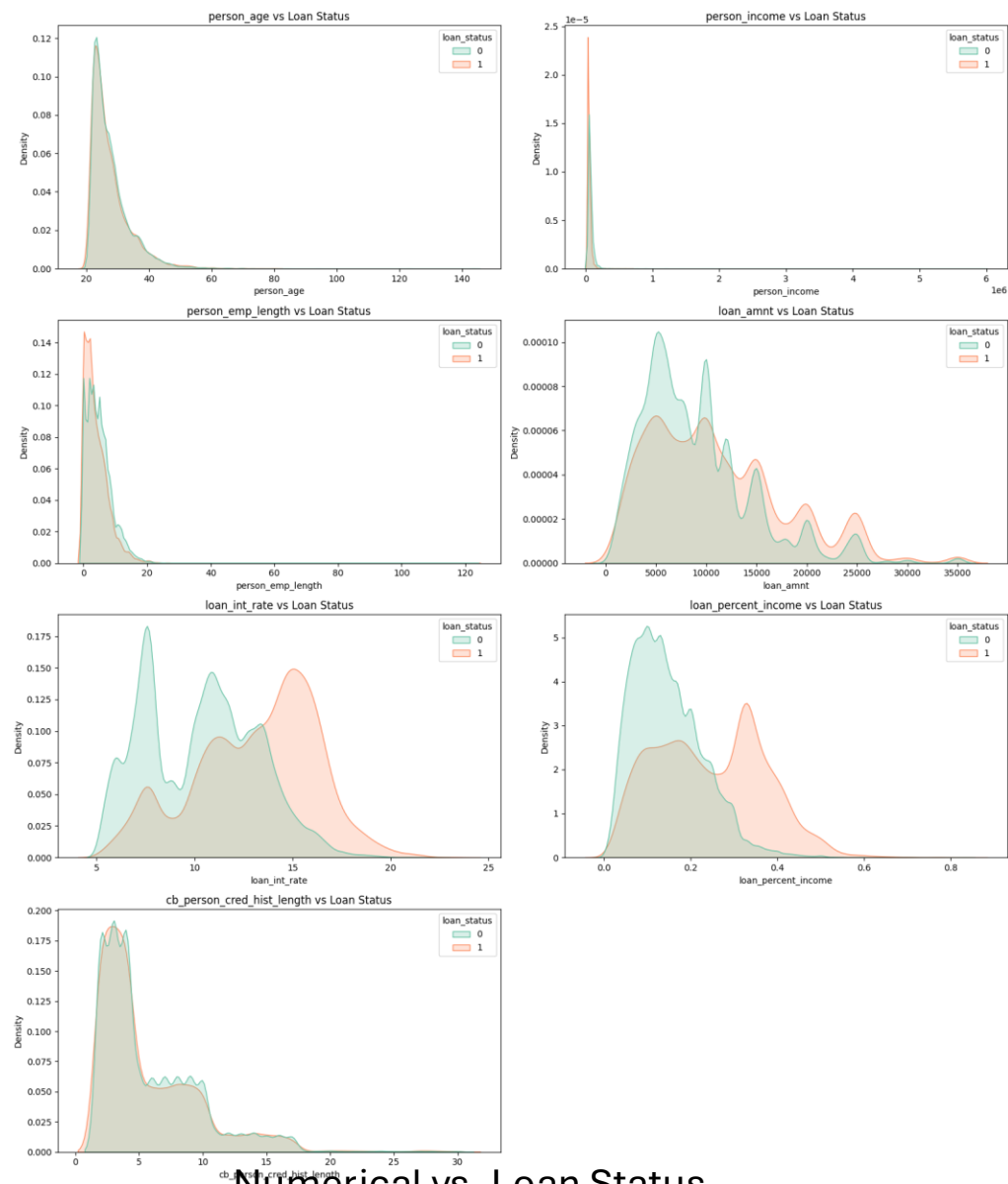
Distribution of numerical variables

2. EDA

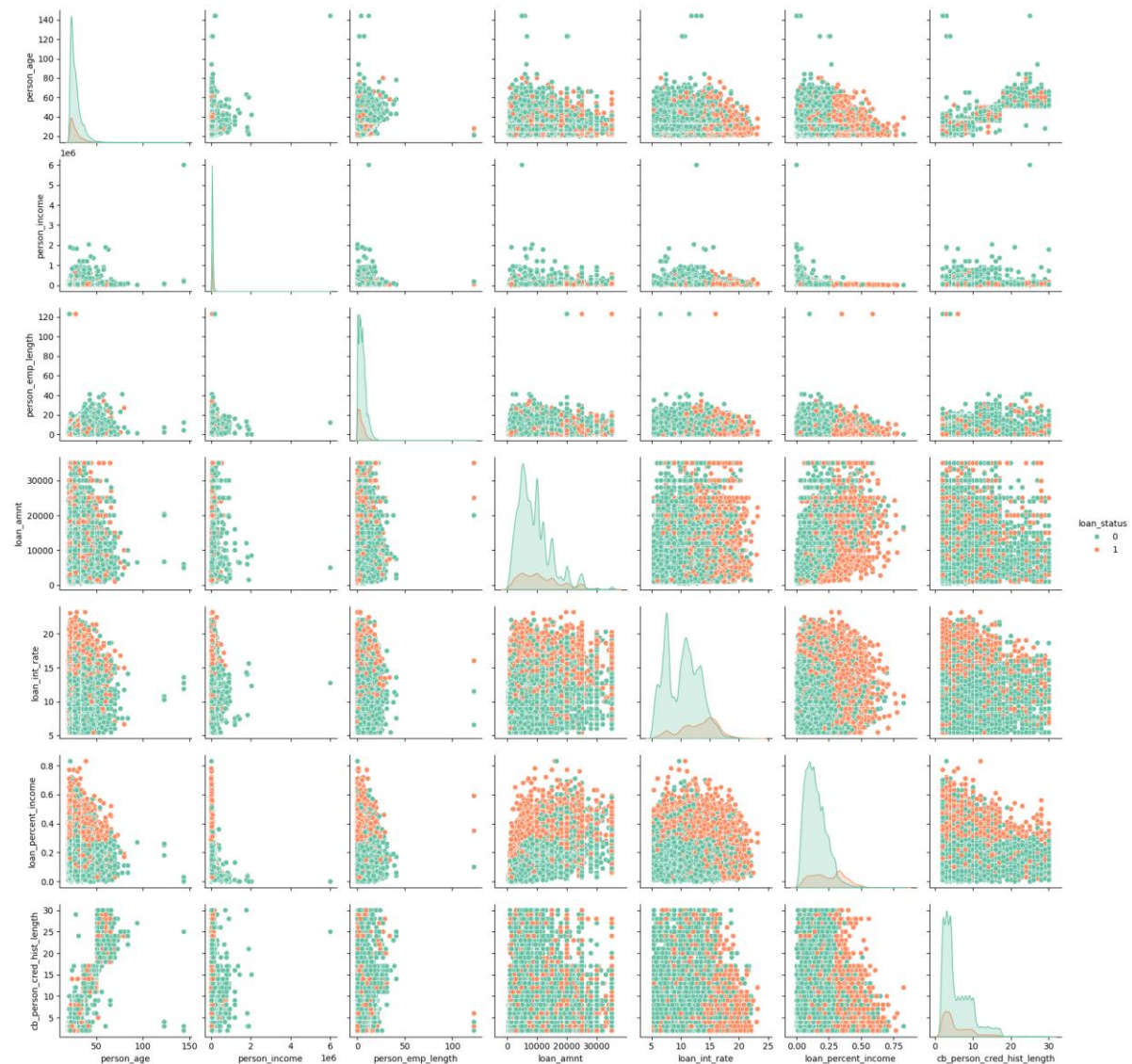


Distribution of categorical variables

2. EDA



Numerical vs. Loan Status



Pairplot of Numerical features

3. Modelling

Predictive Models Used

- Logistic Regression
- Neural Network
- LightBGM
- Random Forest
- XGBoost

Comparison of Tree-Based Models

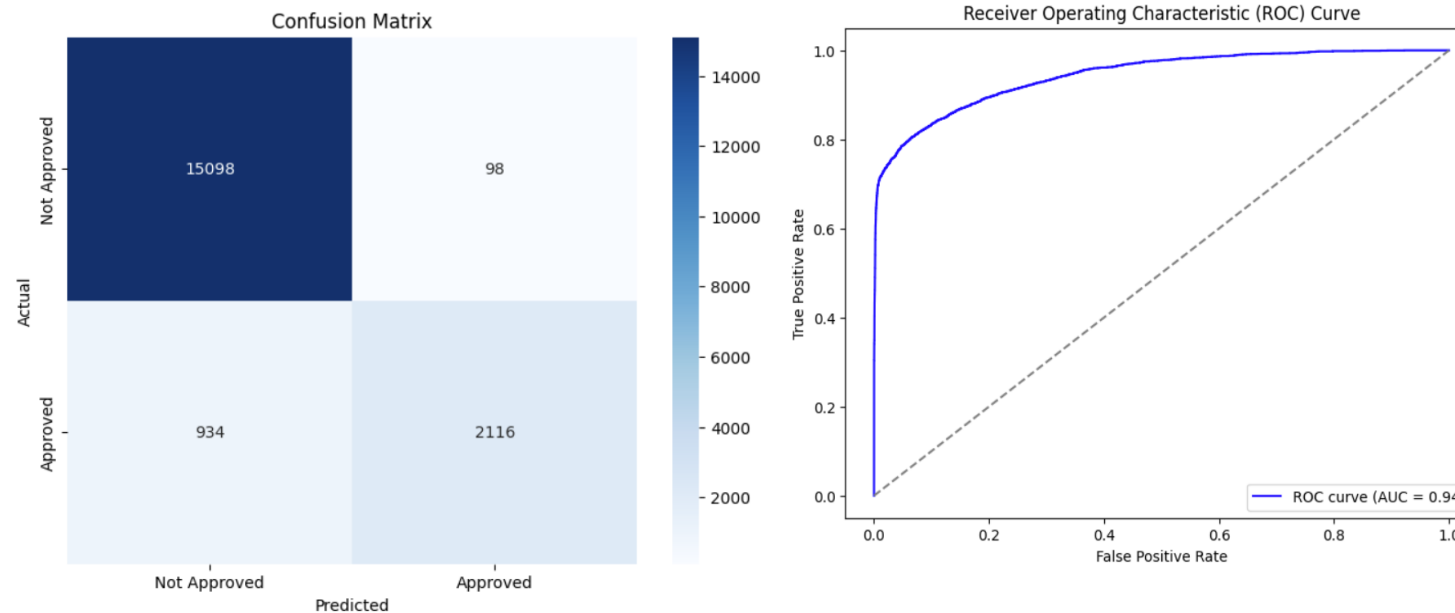
Model	Training Speed	Memory Usage	Handling Large Data
XGBoost	Medium	Medium	Good
Random Forest	Slow	High	Moderate
LightGBM	Fast	Low	Excellent

Table: Comparison of Tree Based Models

3. Modelling

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.8801	0.7178	0.4662	0.5653	0.8791
Logistic Regression (PCA)	0.8805	0.7175	0.4705	0.5683	0.8791
LGBM	0.9032	0.6661	0.8433	0.7443	0.9528
LGBM (PCA)	0.8719	0.5853	0.8007	0.6763	0.8434
Random Forest	0.9431	0.9447	0.7003	0.8044	0.8461
Random Forest (PCA)	0.9258	0.8572	0.6672	0.7504	0.8225
XGBoost	0.9434	0.9557	0.6938	0.8040	0.9436
XGBoost (PCA)	0.9282	0.8699	0.6708	0.7575	0.9267
Neural Network	0.9332	0.9208	0.6529	0.7633	0.9129

Table: Performance Metrics of Different Models



Performance of XGBoost

4. Conclusion / Cost Feasibility

Cost Feasibility

Function	Feature	Cost
Data Collection & Storage	Cloud services, database management	\$50,000
Model Development & Testing	Data scientists, ML engineers	\$100,000
Infrastructure & Security	Cloud computing, Regulatory compliance	\$75,000
Integration with banking systems	API development, fintech partnerships	\$50,000

Total Estimated Costs: \$275,000 ~ \$300,000

Benefits: 30% increase in approval efficiency and up to 20% reduction in loan defaults